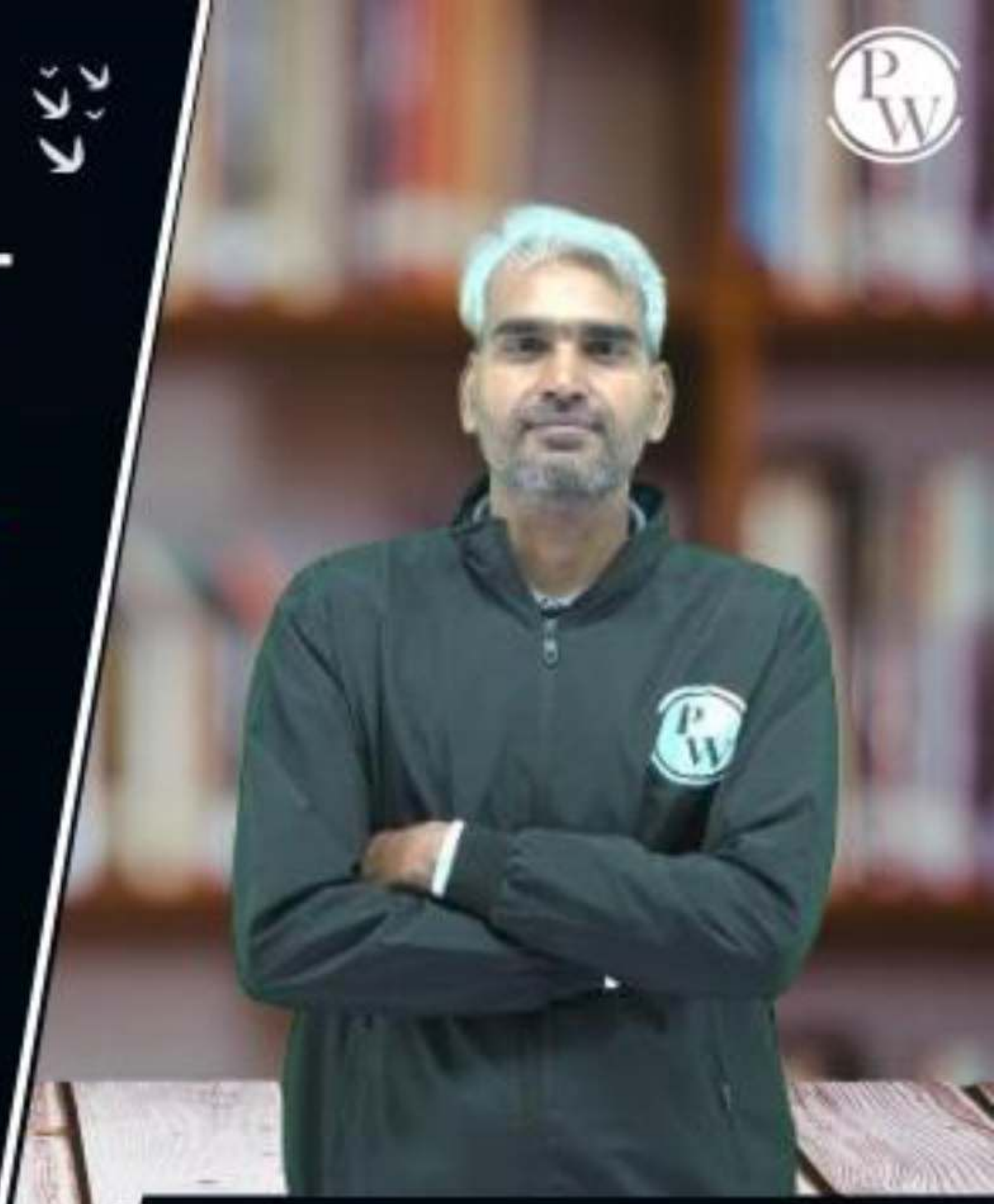
Computer Science & IT

Database Management System

Entity Relationship Model
&
Integrity constraints

Lecture No. 01



By- Vishal Sir

Recap of Previous Lecture

Topic B+ Tree insertion

Topic B+ Tree deletion

Slide

Topics to be Covered

Topic ER model & ER diagram

Topic Terminologies in ER diagram

Slide

Deletion from B+ tree

Topic : Deletion in B+ tree

While deleting a key from B+ tree there are two different cases.

Case 1 : When key to be deleted is present only at a leaf node.

Case 2 : When key to be deleted is present in a leaf node as well as in an internal node. (Keys in the internal node are just dummy keys)

Topic : Deletion in B+ tree

Case 1 : When key to be deleted is present only at a leaf node.

* 1. If leaf node contains more than the minimum number of keys required, then simply delete the key.
* 2. If leaf node contains exact minimum number of keys required, then delete the key and borrow (re-distribute) from the immediate siblings if they can help (left then right order). {when we borrow from the sibling we will have to update the key in the Parent node such that search property is preserved}
3. If siblings can not help, then merge with one of its sibling. Because of this merging if parent node becomes deficient then parent node will borrow from its immediate siblings (if possible) or, merge with one of its siblings (When siblings can not help), it can go up to root node.

Topic : Deletion in B+ tree

Case 2 : When key to be deleted is present in a leaf node as well as in an internal node. (we need to delete the key from both places)

1. If leaf node contains more than the minimum number of keys required, then simply delete the key from leaf node as well as from internal node. Because of deletion from internal node an empty space will get created, fill that empty space of internal node with in-order successor (or predecessor)

When key in internal node is present in right sub-tree

When key in an internal must be present in left sub-tree

Topic : Deletion in B+ tree

Case 2 : When key to be deleted is present in a leaf node as well as in an internal node. (we need to delete the key from both places)

2. If leaf node contains exact minimum number of keys required and any of the immediate sibling can help, then simply delete the key from leaf node as well as from internal node, borrow the key from the sibling and fill that empty space of internal node with in-order successor (or predecessor)

Topic : Deletion in B+ tree

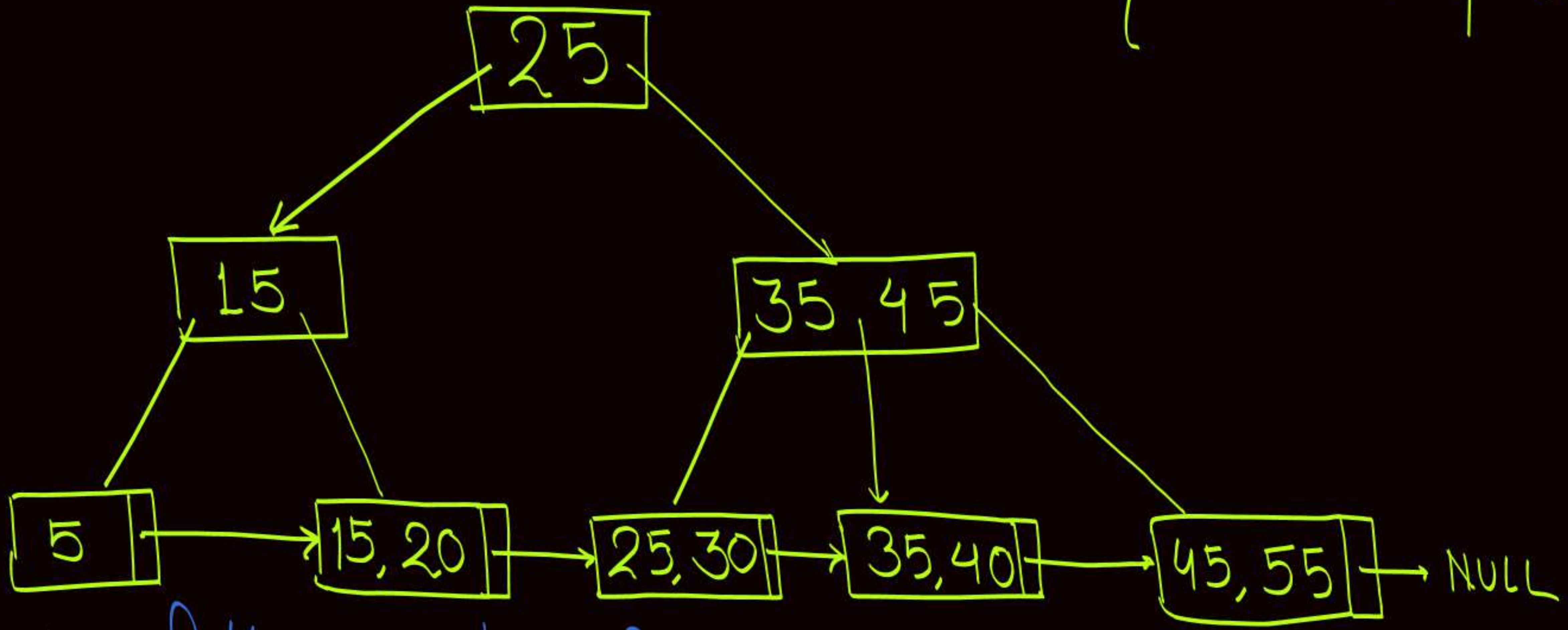
Case 2 : When key to be deleted is present in a leaf node as well as in an internal node. (we need to delete the key from both places)

3. If leaf node contains exact minimum number of keys required and none of the immediate sibling can help, then simply delete the key from leaf node as well as from internal node and merge the leaf node with one of its sibling, it may create deficiency in the parent node therefore repeat the process for parent node as usual.

If empty space created in internal node is not handled while merging the parent node then fill it with in-order successor (or predecessor).

H.W. :-

Q:- Consider the following B+ tree { let order of internal node = 3, Order of leaf node = 2 }



Delete the following keys from B+ tree in the same order

40, 5, 45, 35, 25, 55

Delete: ~~40~~, 5, 45, 35, 25, 55

order of,
internal node = 3
leaf node = 2

delete 40

Min no. of keys a non-root internal node must have $= \lceil \frac{3}{2} \rceil - 1 = 1$

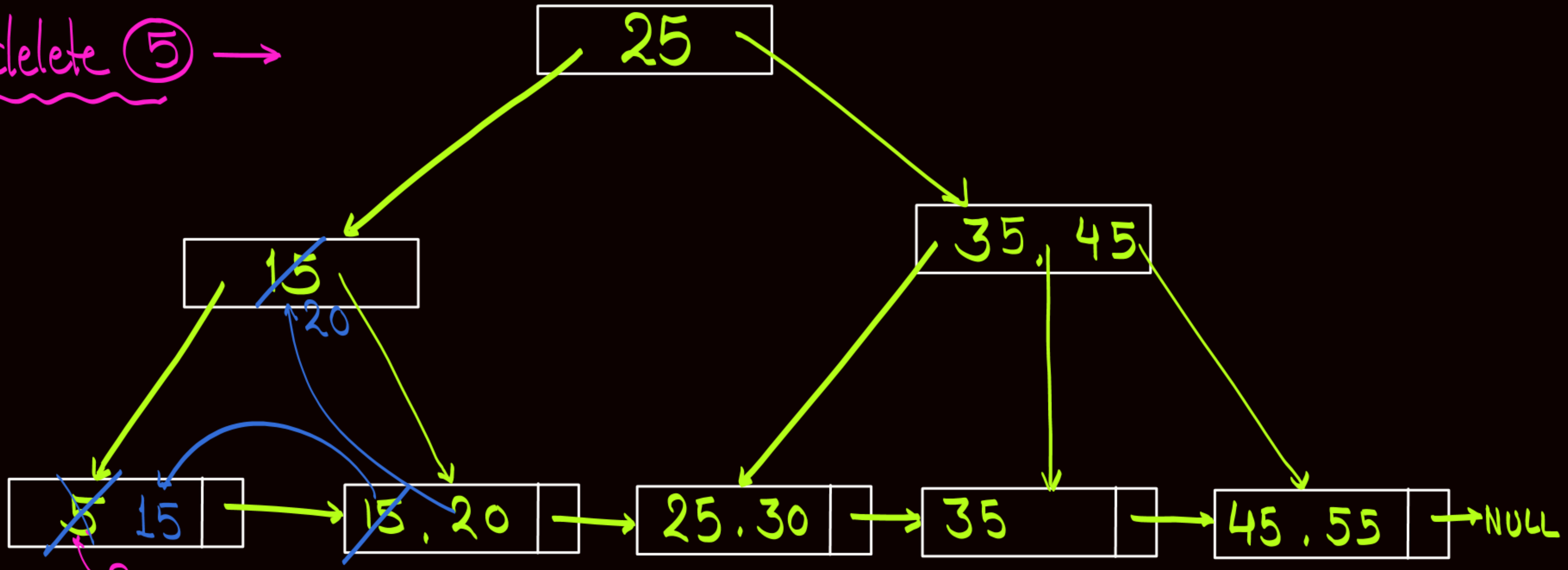
Min. no. of keys a non-root leaf node must have $= \lceil \frac{2}{2} \rceil = 1$

Present only in leaf node & node Contains more than the minimum no. of keys required

∴ Simply delete it

Delete: ~~40~~, 5, 45, 35, 25, 55

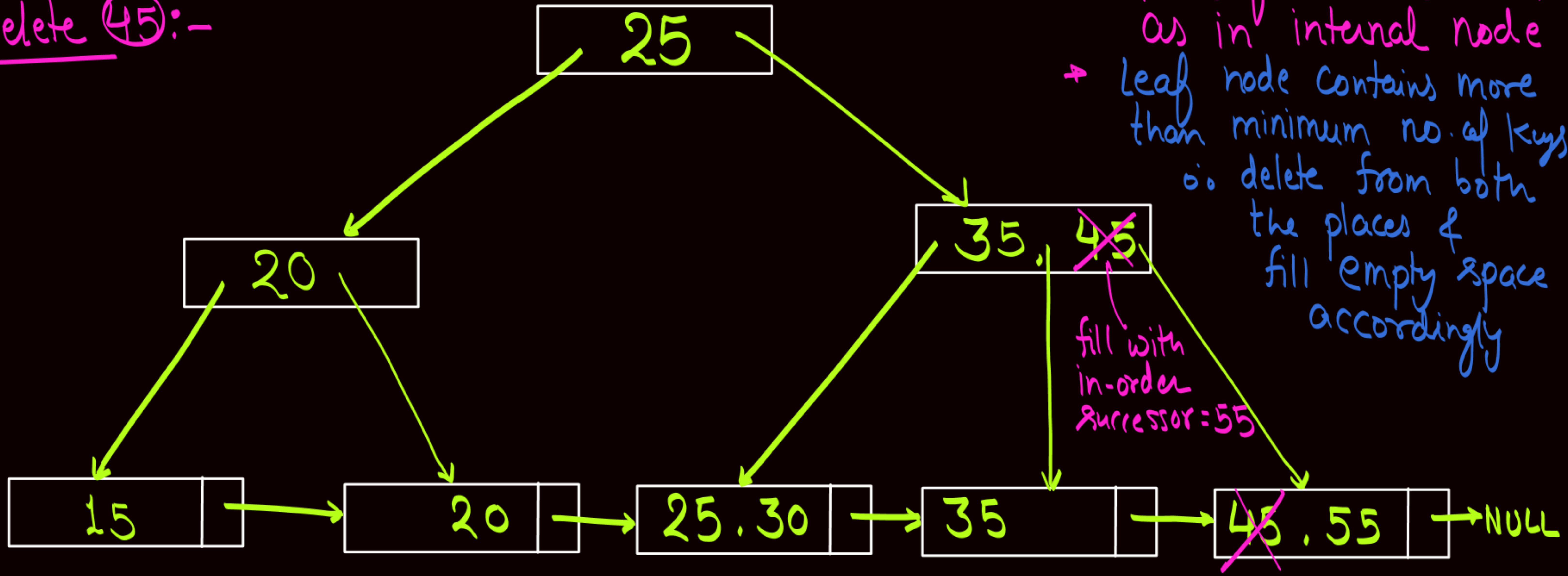
delete 5 →



- → Present only in leaf node
- → Node Contains exact minimum number of keys, but right sibling can help.
- → ∴ delete & borrow from right sibling {and update key in parent node if needed}

Delete: ~~40~~, ~~5~~, 45, 35, 25, 55

delete 45 :-



* 45 is present in leaf node as well as in internal node

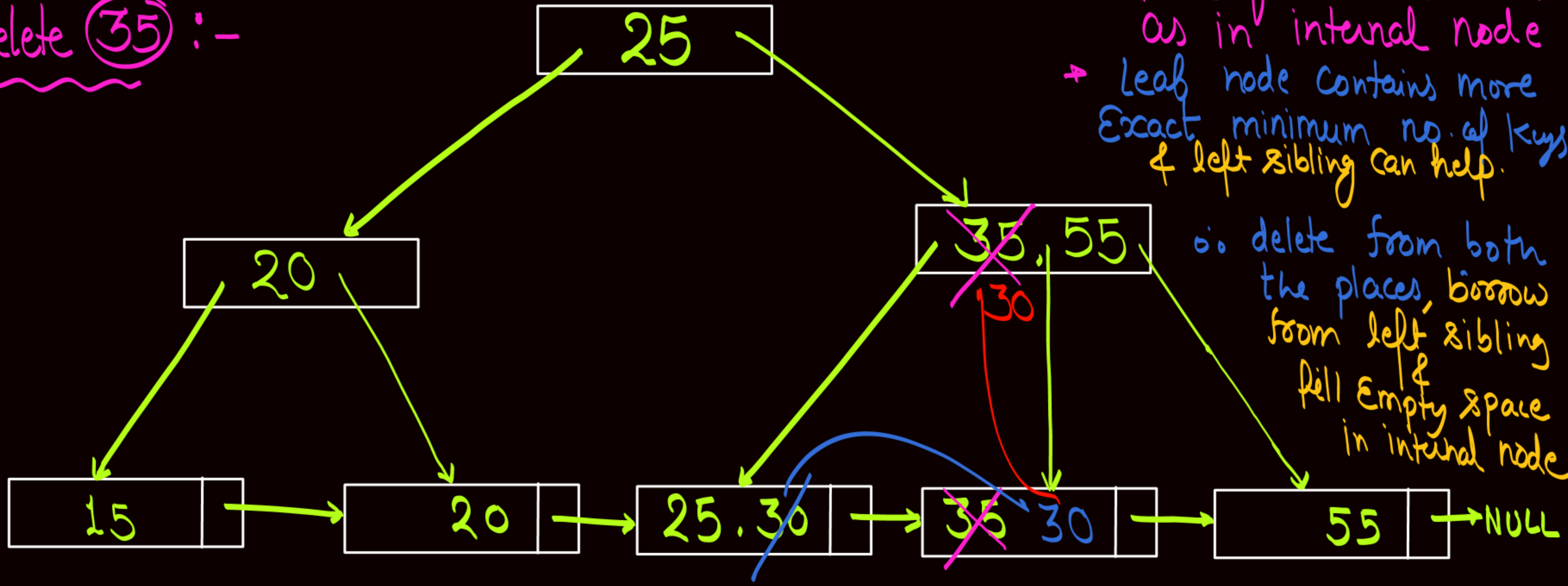
* leaf node contains more than minimum no. of keys
∴ delete from both the places & fill empty space accordingly

Delete: ~~40~~, ~~5~~, ~~45~~, 35, 25, 55

delete 35 :-

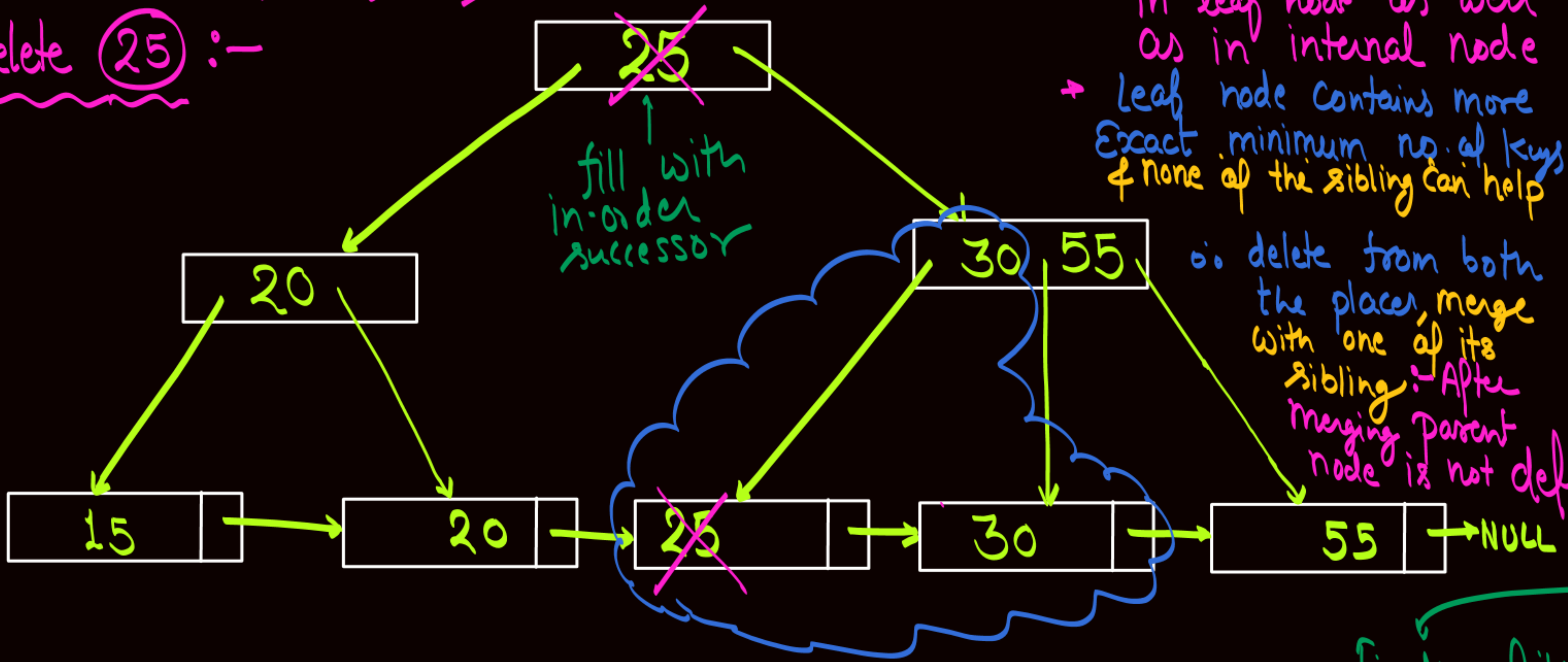
* 35 is present in leaf node as well as in internal node
* leaf node contains more Exact minimum no. of keys & left sibling can help.

∴ delete from both the places, borrow from left sibling & fill Empty space in internal node



Delete: ~~40~~, ~~5~~, ~~45~~, ~~35~~, 25, 55

delete 25 :-



* 25 is present in leaf node as well as in internal node

→ leaf node contains more Exact minimum no. of keys & none of the sibling can help

∴ delete from both the places, merge with one of its sibling :- After merging parent node is not deficient

Finally fill the Empty space with in-order successor

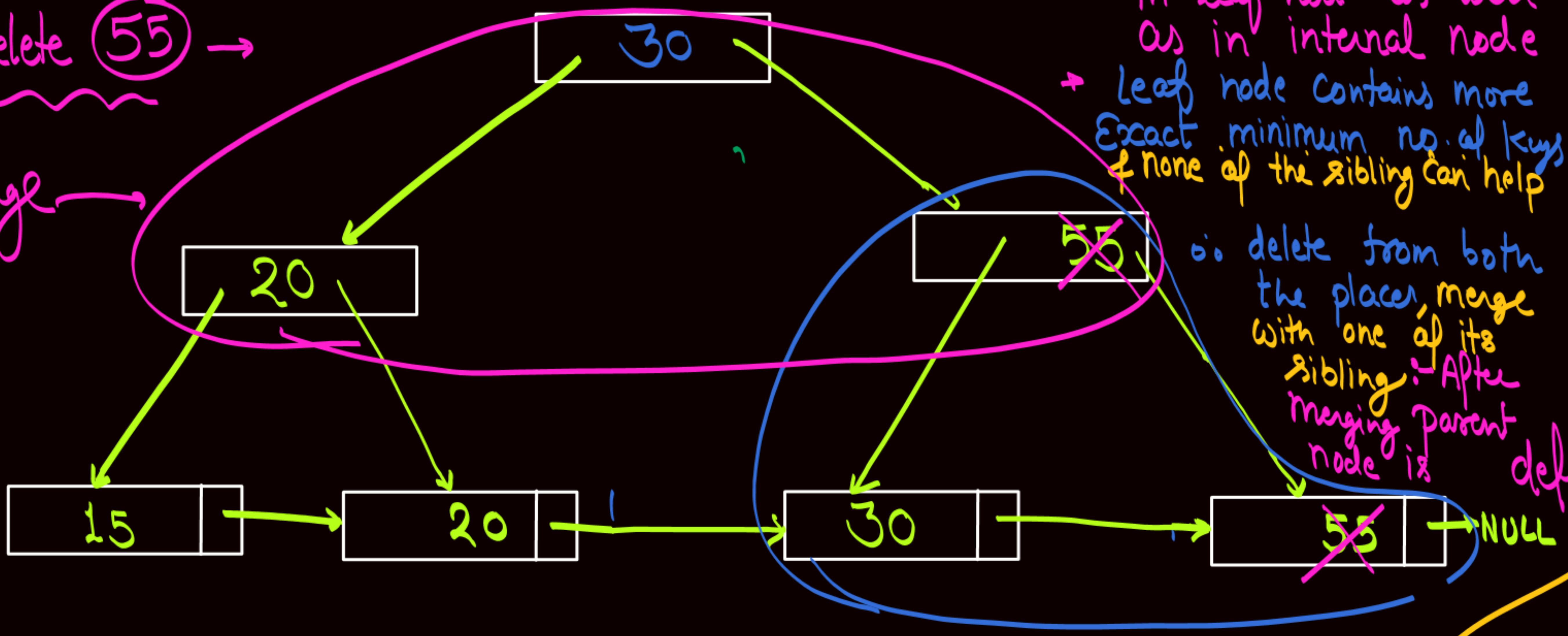
Delete: ~~40~~, ~~5~~, ~~45~~, ~~35~~, ~~25~~, 55

delete 55 →

* 55 is present in leaf node as well as in internal node
* leaf node contains more Exact minimum no. of keys & none of the sibling can help

∴ delete from both the places, merge with one of its sibling. After merging parent node is deficient

Merge →



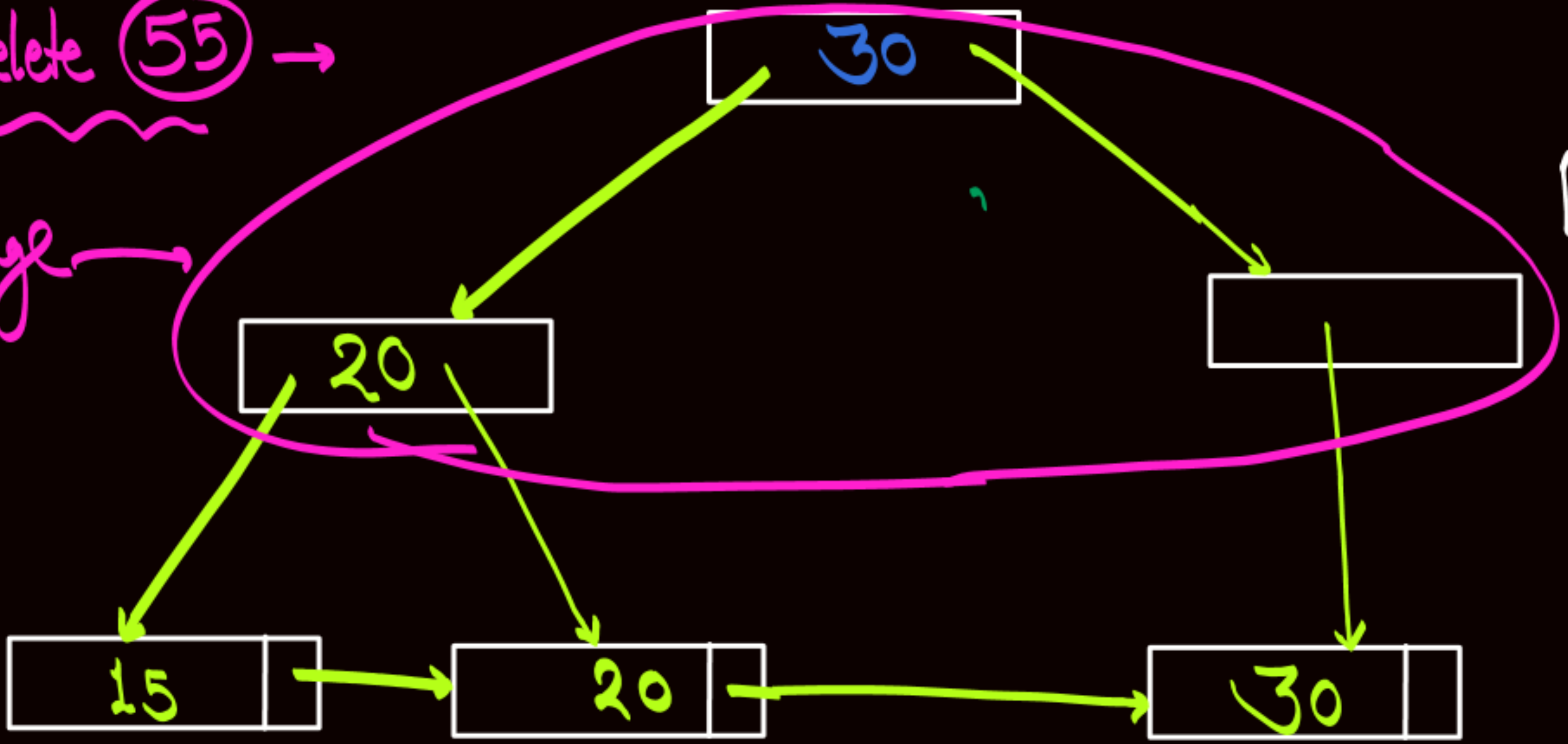
Sibling of the parent node can not help.

∴ Merge with sibling

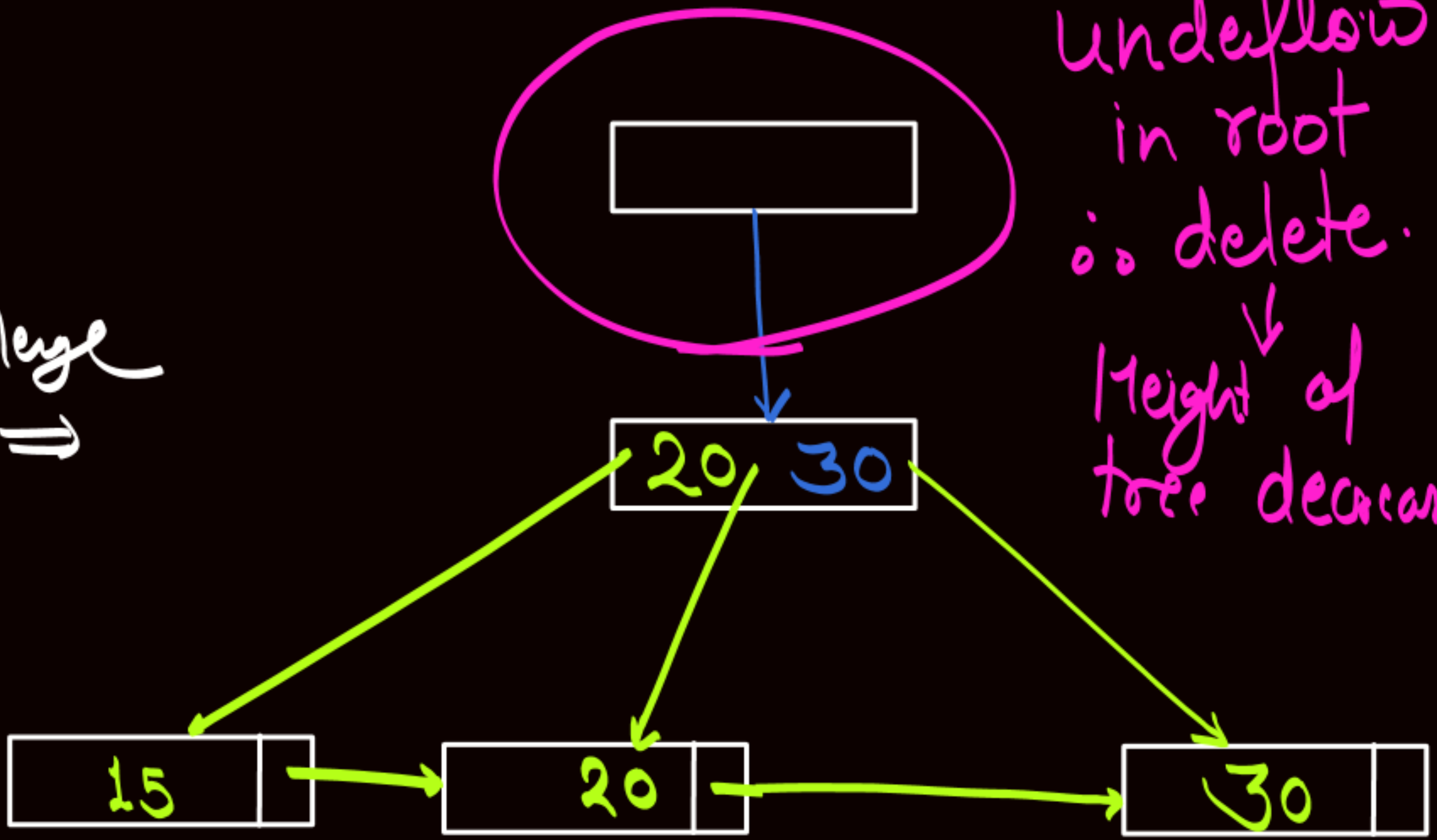
Delete: ~~40~~, ~~5~~, ~~45~~, ~~35~~, ~~25~~, 55

delete 55 →

Merge →

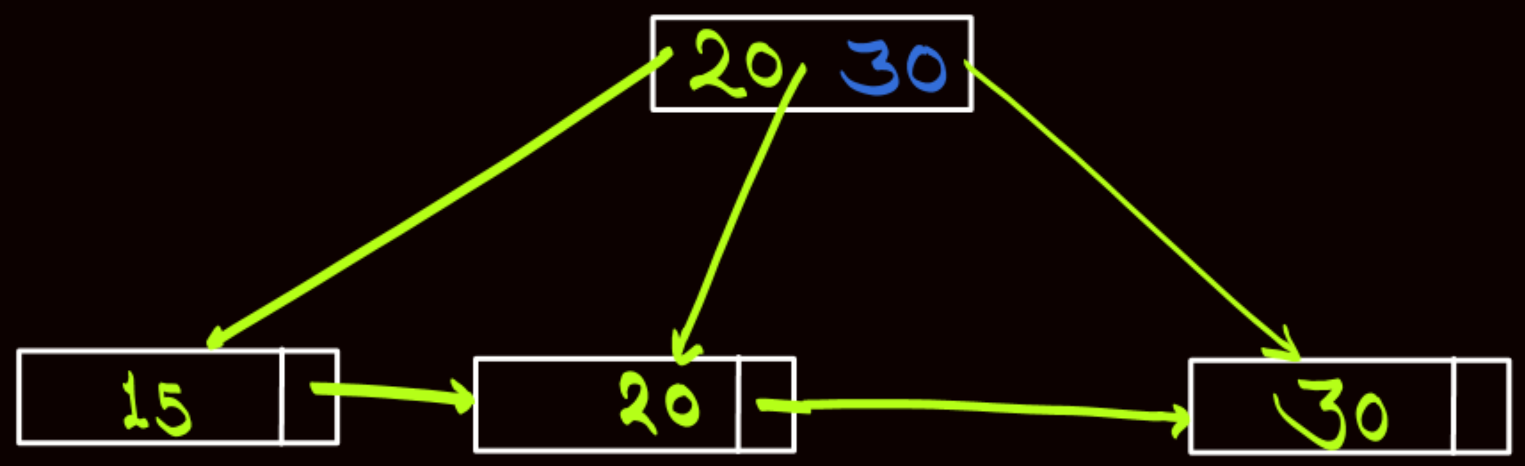


Merge ⇒



underflow in root ∴ delete.
↓
Height of tree decreases

Delete: ~~40~~, ~~5~~, ~~45~~, ~~35~~, ~~25~~, ~~55~~



ER Model

Topic : Entity-Relationship Model (ER-Model)

- ER Model is a high-level data model diagram which defines the conceptual view of the database (OR) that gives the graphical representation of the logical structure of the database.

- Entity-Relationship model is based on objects called entities, and relationship among these entities.

Topic : Entity

- An entity refers to any object having either a physical existence or a conceptual existence. For example, a school, a person, a house, a university, a company or a job.

Topic : Entity Set

An entity set is a collection or set of all entities of a particular entity type at any point in time. "Student"

Roll_no	Name	Age
1	Ram	20
2	Shyam	19
3	Mohan	20
4	Arjun	19

This complete table is referred as "Student Entity Set" and each row represents an "entity".

Topic : Entity Set

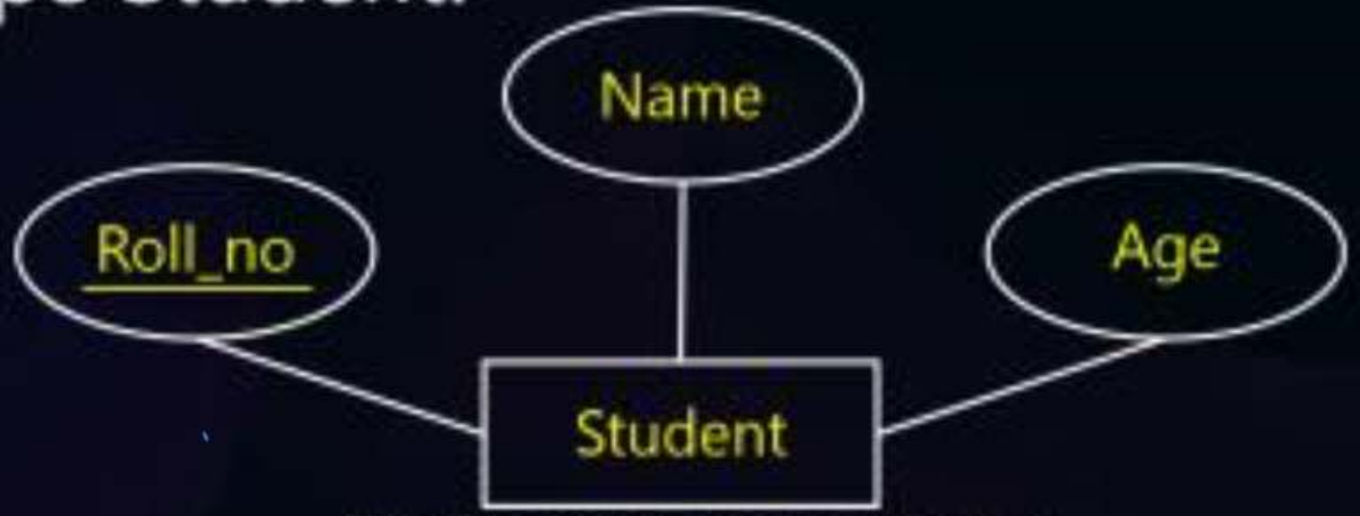
An entity set is represented by a rectangle in ER diagram

eg: $\boxed{\text{Student}}$

Topic : Attribute

- Each entity is identified uniquely by the values of its attributes.
- An attribute in an Entity-Relationship Model describes the properties or characteristics of an entity. It is represented by an oval or ellipse shape in the ER diagram. Every oval shape represents one attribute and is directly connected to its entity set which is rectangle in shape.

For example, roll_no, name, and age are the attributes which define entity type Student.



Entity representation

Roll_no	Name	Age
1	Ram	20
2	Shyam	19
3	Mohan	20
4	Arjun	19

Topic : Types of Attributes

i. Simple attribute

ii. Composite attribute

iii. Single-valued attribute

iv. Multi-valued attribute

v. Key attribute

vi. Derived attribute

vii. Partial key attribute

Topic : Types of Attributes

(i) **Simple Attributes:** An attribute which contains an atomic value and cannot be divided further is called a simple attribute. Example, Roll_no, Sex, etc.

(ii) **Composite attribute:** An attribute which can divided further. It is an attribute which is composed of several components. Example name, address, etc.

(iii) **Single-valued attribute:** An attribute which can have only one value. {eg. Sname}

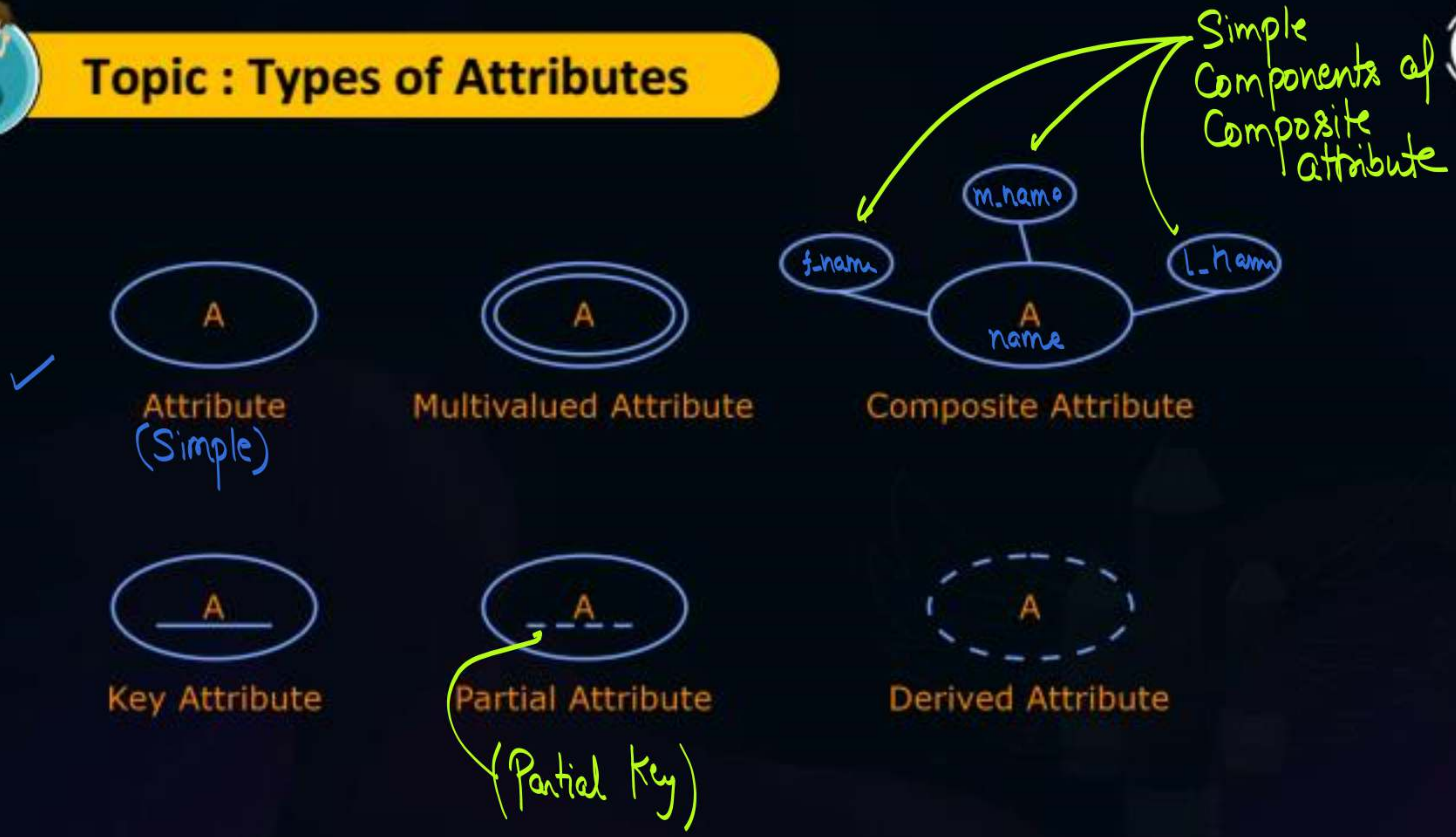
(iv) **Multi-valued attribute** : An attribute which may have multiple values (more than one value) {eg Mob_No. of student}

(v) **Key attribute** : An attribute which can identify all entities uniquely.

(vi) **Derived attribute** : An attribute whose value can be derived or found from any other attribute is known as derived attributes. {eg Age can be obtained if D.O.B is stored}

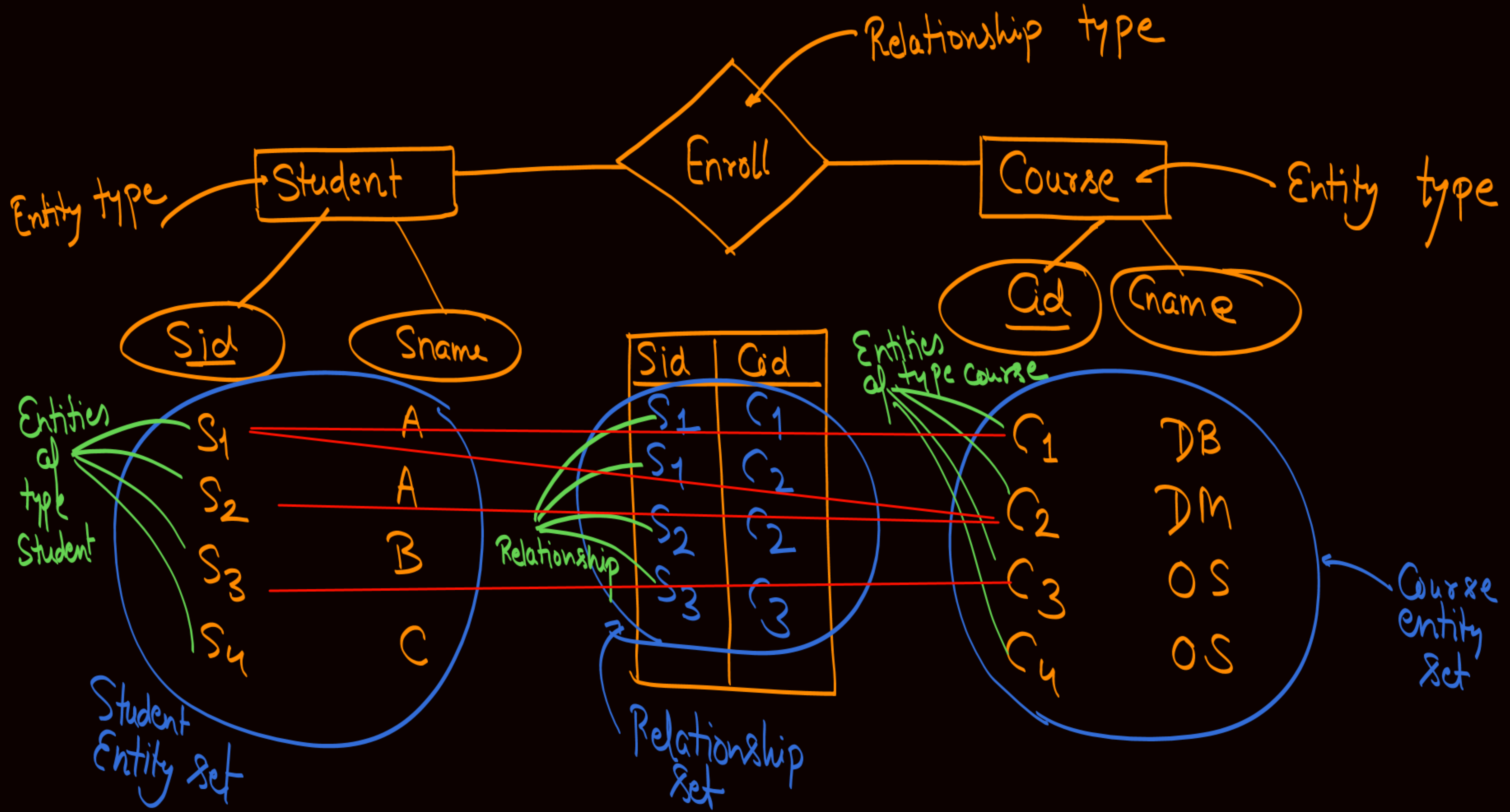
(vii) **Partial key attribute** : An attribute which can identify a group of tuples.

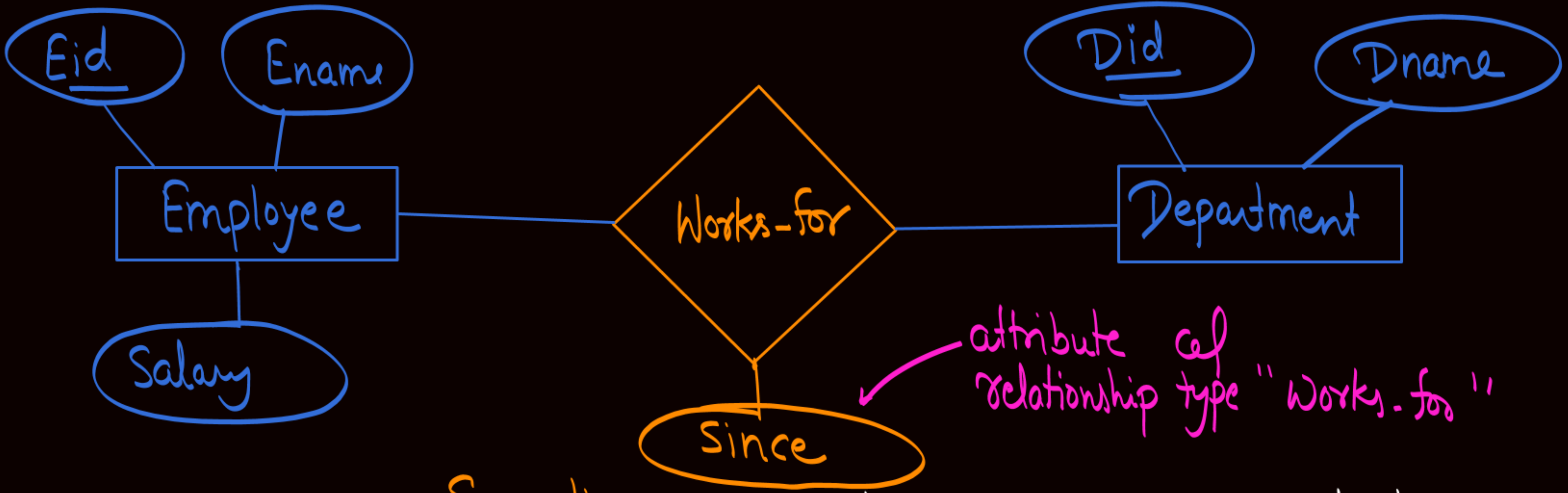
Topic : Types of Attributes



Topic : Relationship in ER model

- A Relationship Type represents the association between entity types. For example, 'Enrolled' is a relationship type that exists between entity type Student and Course. In ER diagram, the relationship type is represented by a diamond and connecting the entities with lines.


Relationship type
Enroll
Student
Course
Entity type
Entity type
Sid
Sname
Cid
Cname
Entities of type Student
S1 A
S2 A
S3 B
S4 C
Sid Cid
S1 C1
S1 C2
S2 C2
S3 C3
Relationship
Entities of type course
C1 DB
C2 DM
C3 OS
C4 OS
Student Entity set
Relationship set
Course entity set



attribute of relationship type "works-for"

it will be used to specify that since when the Employee is associated with that department

Some times relationship type may have their own attributes

{ Attributes of "relationship type" are called "descriptive attributes"

Note :- It is not necessary for Every relationship type to have their own attributes.

Topic : Descriptive attribute

Descriptive attributes are use to provide the information related to relationship set.

Topic : Degree of a Relationship

- The number of different entity sets participating in a relationship is called the degree of a relationship.

Degree of a relationship = Number of entity sets participating in a relationship

Topic : Types of Relationship

On the basis of degree of a relationship set, a relationship set can be classified into the following types-

i. **Unary Relationship:** Unary relationship set is a relationship set where only one entity set participates in a relationship set.

ii. **Binary Relationship:** Binary relationship set is a relationship set where two entity sets participate in a relationship set.

iii. **Ternary Relationship:** Ternary relationship set is a relationship set where three entity sets participate in a relationship set.

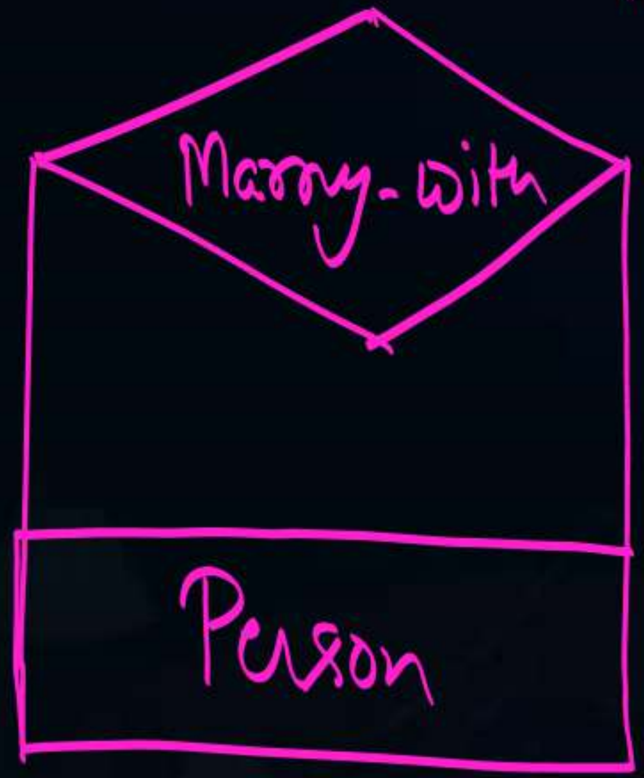
iv. **N-ary Relationship:** N-ary relationship set is a relationship set where 'N' entity sets participate in a relationship set.

Topic : Unary Relationship

Only one Entity type "Person" is participating in relationship type "Marry-with"

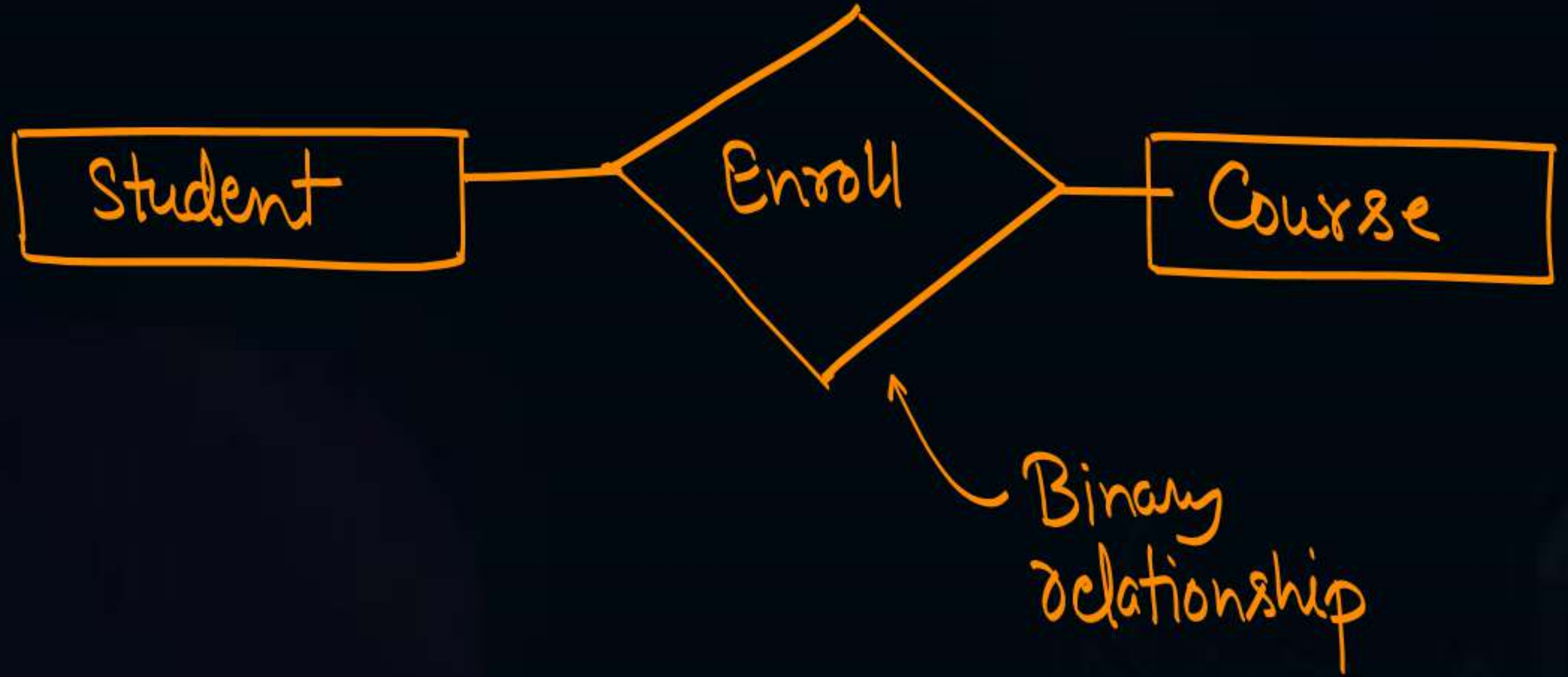
∴ Marry with a Unary Relation-ship

Unary relationship

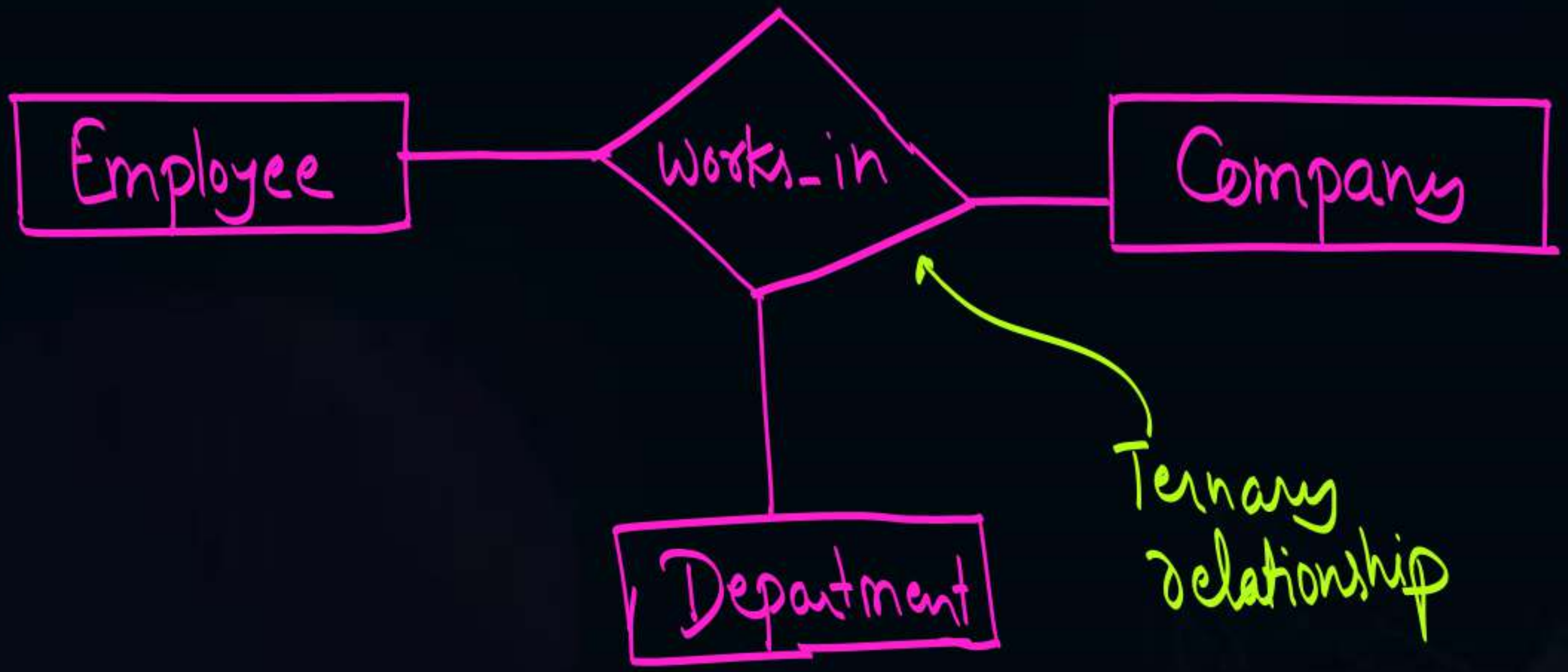


Topic : Binary Relationship

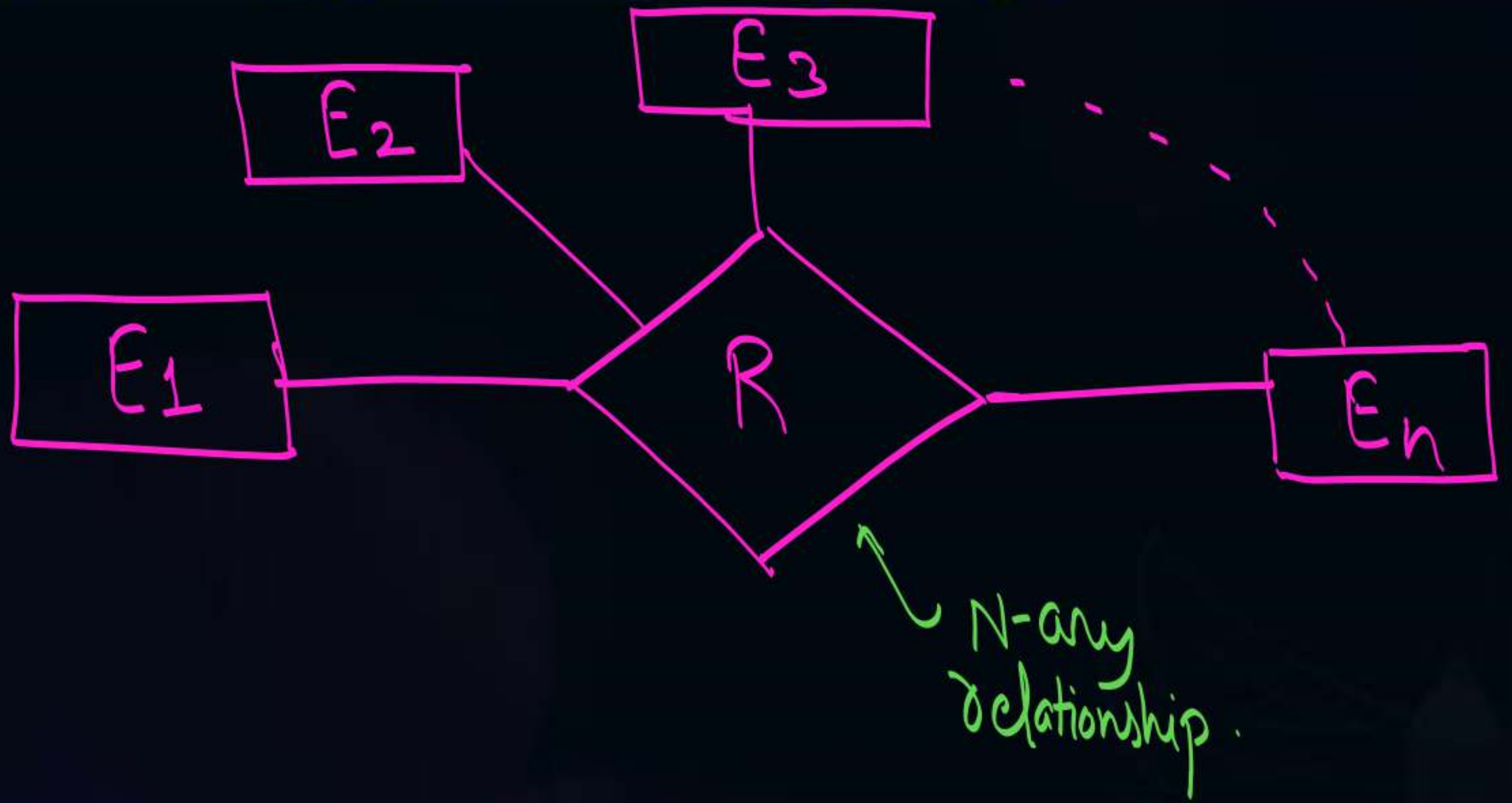
eg:



Topic : Ternary Relationship



Topic : N-ary Relationship



Topic : Participation constraints

* There are two types of participation constraints :

1. Total participation
2. Partial participation

Topic : Total participation

- If every entity of an entity set participate in the relationship set, then that entity set is said to have total participation.
- Total participation of an entity set in a relationship set is denoted by double line

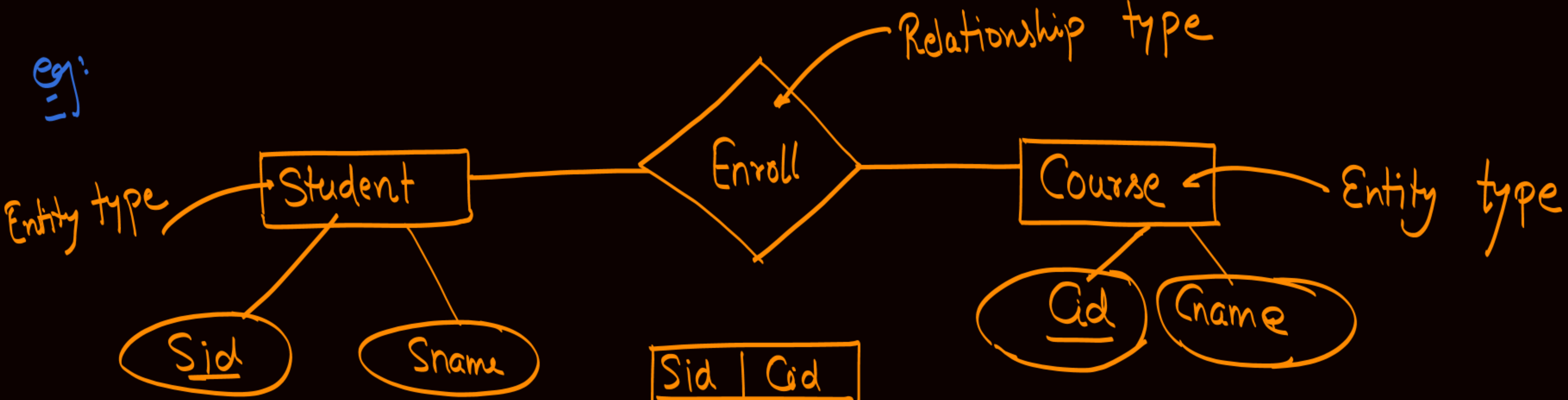
it represent total participation

it represent that Every Entity of Entity Set E_2 Will Participate in relationship R

Topic : Partial participation

- If at least one entity of entity set is not participating in the relationship set, then that entity set is said to have partial participation.
- Partial participation of an entity set in a relationship set is denoted by single line.

eg:



Sid	Sname
S_1	A
S_2	A
S_3	B
S_4	C

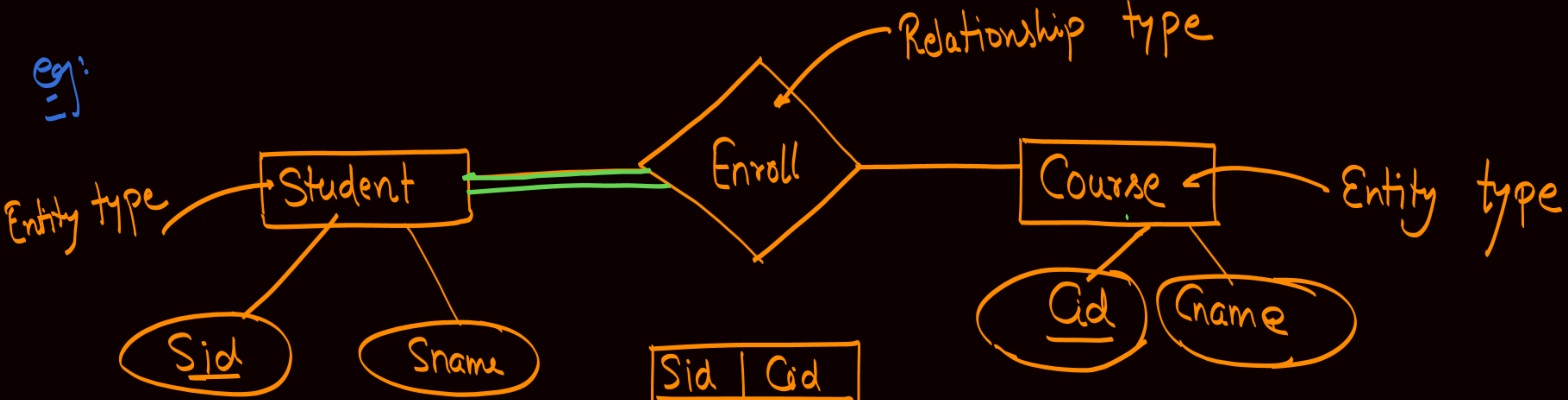
S_4 → Not participating
$\therefore$ Partial Participation of Student

Sid	Cid
S_1	C_1
S_1	C_2
S_2	C_2
S_3	C_3

Cid	Cname
C_1	DB
C_2	DM
C_3	OS
C_4	OS

C_4 → Not Participating
$\therefore$ Partial Participation of Course

eg:



S_1	A
S_2	A
S_3	B
S_4	C

Sid	Cid
S_1	C_1
S_1	C_2
S_2	C_2
S_3	C_3
S_4	C_3

C_1	DB
C_2	DM
C_3	OS
C_4	OS

Every Entity of Student Entity set is Participating
∴ Total Participation

Not Participating
∴ Partial Participation of Course

2 mins Summary

Topic ER model & ER diagram

Topic Terminologies in ER diagram

Slide

THAK - YOU

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